
MODEL SELECTION AND VALIDATION REPORT (boreal_biomass model)

OBSERVATIONS:1023

SANITY: warning - sanity check failed for the following models:
spatiotemporal_random_fields_iid_with_anisotropy

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1_with_anisotropy

MODEL SUMMARY (boreal_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	7.18	2.25
kde_fric	0.10	0.03
kde_feve	-0.14	0.03
tric	0.10	0.03
teve	-0.47	0.03
sbt_mean	0.05	0.08
sic_mean	0.03	0.04
sbt_sd	0.09	0.03
sst_sd	-0.05	0.06
log_chla_mean	-0.04	0.03
depth	-0.10	0.04

Dispersion parameter: 0.67

Spatiotemporal AR1 correlation (rho): 0.99

Matérn anisotropic range (spatiotemporal): 617.9 to 2487.3 at 108 deg.

Spatiotemporal marginal AR1 SD: 1.50

REML criterion at convergence: 8382.789

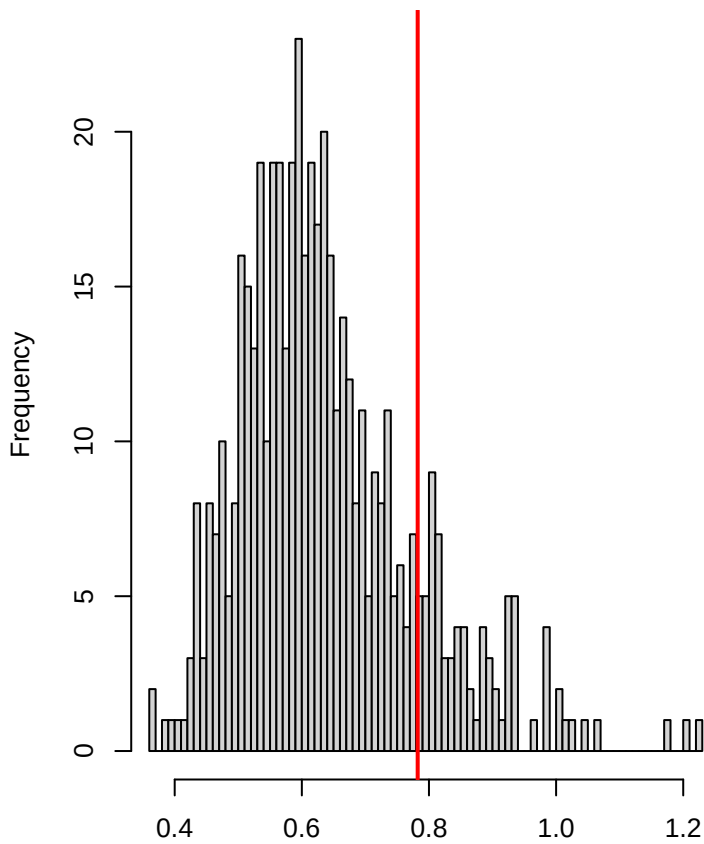
See ?tidy.sdmTMB to extract these values as a data frame.

See ?plot_anisotropy to plot the anisotropic range.

SELECTION STATISTICS

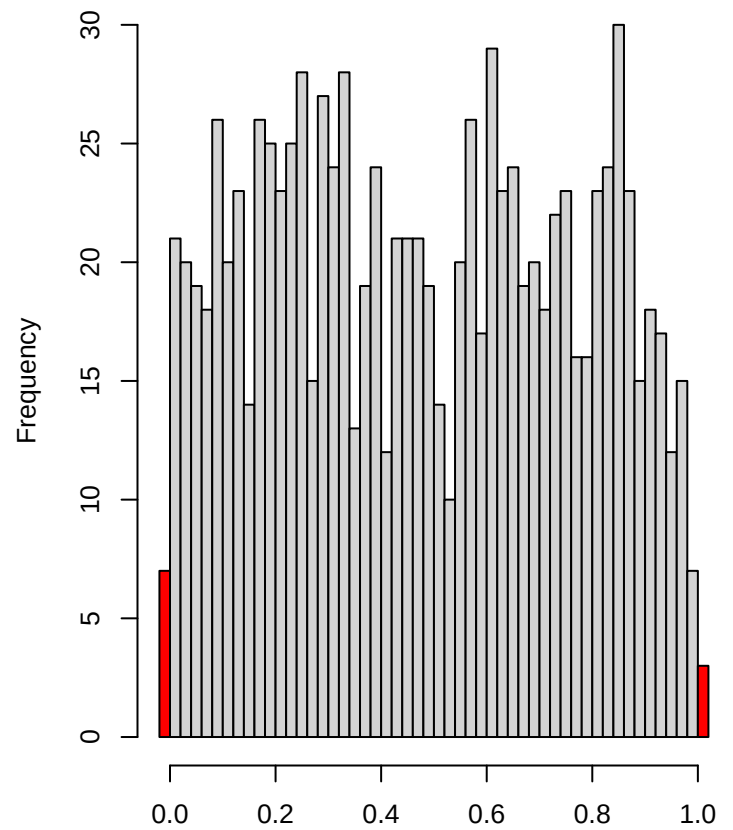
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1_with_anisotropy	17	16799.58	2.520032
spatiotemporal_random_fields_ar1	15	16802.10	47.692605
spatiotemporal_random_fields_iid	14	16849.79	1.201238
spatial_random_fields_with_anisotropy	16	16850.99	1.845286
spatial_random_fields	14	16852.84	5.524590
spatiotemporal_random_fields_iid_with_anisotropy	16	16858.36	56.382729
error_only	12	16914.75	NA

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated



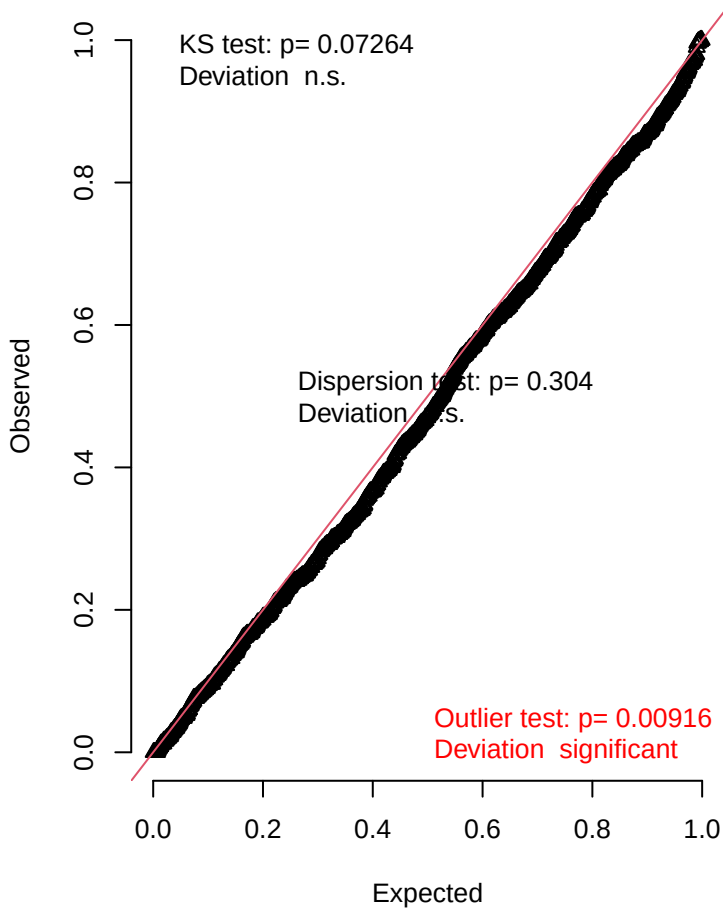
Simulated values, red line = fitted model. p-value (two.sided) =

Outlier test significant

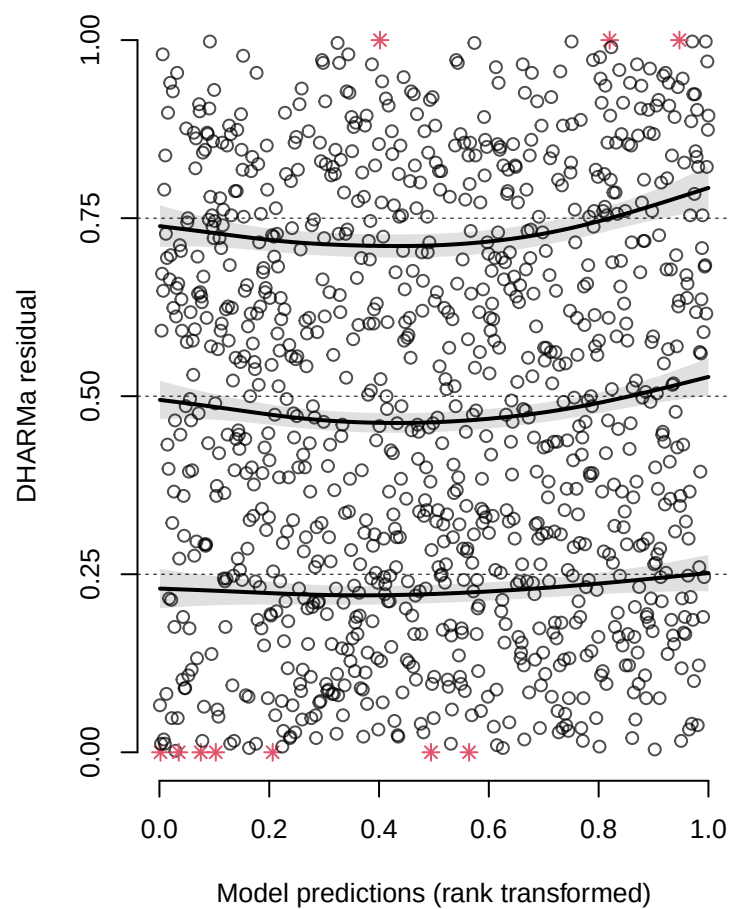


Residuals (outliers are marked red)

QQ plot residuals



DHARMA residual vs. predicted
No significant problems detected



DHARMa Moran's I test for distance-based autocorrelation

Moran's I = 0.013 | H1: Distance-based autocorrelation | p-value = 0.194

